MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR

EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCFSC300 - FORENSIC BALLISTICS

(2024 ADMISSION ONWARDS)

SCHEME OF VALUATION

Duration: 1 hour 30 minutes **Maximum Marks: 50**

General instructions: Award marks for equivalent scientifically correct answers. The points below are indicative, not exhaustive. For questions with multiple parts, award marks proportionately. Do not exceed the maximum mark for each question.

Part A – Short Answer Type Questions

Answer any 10 questions. Each question carries 2 marks. Maximum: 20 marks.

1. What is a double-action firearm? (2 marks)

- Trigger pull both cocks and releases the hammer/striker — 1 mark
- This allows firing without manually cocking before each shot — 1 mark

2. What are shock waves associated with a projectile? (2 marks)

- Pressure disturbances produced as a projectile travels rapidly through air — 1 mark
- At supersonic speed, these form a characteristic shock/conical wave — 1 mark

3. What is meant by matching crime bullets with test bullets? (2 marks)

- Comparison of a recovered crime bullet with a test-fired bullet — 1 mark
- Microscopic comparison of rifling marks/striations to assess common firearm origin — 1 mark

4. What is a primer composition? (2 marks)

- The impact-sensitive chemical mixture in a cartridge primer — 1 mark
- Ignites when struck and initiates the propellant charge; examples of components may be credited — 1 mark

5. Mention any two factors affecting pressure inside a firearm barrel. (2 marks)

- Any two: propellant type/quantity and burning rate — 1 mark
- Any other valid factor: bullet fit/weight, case capacity, barrel dimensions, temperature, or obstruction — 1 mark

6. What is the Walker test? (2 marks)

- A chemical test historically used to detect nitrite residues around a bullet hole — 1 mark
- May help estimate firing distance; it is presumptive and must be interpreted with controls/limitations — 1 mark

7. Explain the forensic significance of bullet striations. (2 marks)

- Striations are microscopic marks imparted by the barrel during passage — 1 mark
- Their pattern can support comparison of a questioned bullet with test bullets from a suspected firearm — 1 mark

8. A bullet loses its gyroscopic stability during flight. What effect would this have on its motion? (2 marks)

- The bullet may yaw or tumble rather than remain point-forward — 1 mark
- This can increase drag and reduce accuracy/range, and may affect impact behaviour — 1 mark

9. What are the characteristics of a contact shot? (2 marks)

- Muzzle is in contact with or very close to the skin/target — 1 mark
- Possible muzzle imprint, soot/gases within wound, searing/tearing, and dense residue; findings vary by weapon and site — 1 mark

10. What is the scope of forensic ballistics in criminal investigation? (2 marks)

- Examination of firearms, ammunition, bullets, cartridge cases and firing effects — 1 mark
- Comparison/linkage, range/direction assessment and reconstruction of shooting events — 1 mark

11. A bullet strikes a hard surface at an angle and changes its direction. Identify and explain the phenomenon involved. (2 marks)

- Ricochet — 1 mark
- Projectile deflects after impact with a surface; deformation, surface marks and altered trajectory may occur — 1 mark

12. Differentiate between muzzle flash and muzzle blast. (2 marks)

- Muzzle flash: visible light from hot gases/combustion products at the muzzle — 1 mark
- Muzzle blast: pressure wave/noise caused by rapidly expanding gases — 1 mark

Part B – Short Essay Type Questions

Answer any 5 questions. Each question carries 4 marks. Maximum: 20 marks.

13. Explain the role of propellants in ammunition and mention their important characteristics. (4 marks)

- Generate hot gases on combustion to propel the projectile — 1 mark
- Controlled, rapid gas generation and suitable pressure curve — 1 mark
- Characteristics: chemical stability and predictable burning rate — 1 mark
- Adequate energy with manageable residue/corrosion and safe storage; any valid point — 1 mark

14. Explain the role of the National Integrated Ballistic Information Network (NIBIN) in linking firearm-related incidents. (4 marks)

- Digital system/database for ballistic evidence, especially cartridge-case images — 1 mark
- Captures and compares characteristic breech-face/firing-pin and other marks — 1 mark
- Generates investigative leads linking shootings or evidence items — 1 mark
- Candidate associations require confirmation by trained examiner; database result alone is not final identification — 1 mark

15. Describe the important identifying features of a 12-bore firearm and a .315 firearm. (4 marks)

- 12-bore: shotgun gauge designation; commonly smoothbore and fires shotshells, though configurations vary — 1 mark
- 12-bore cartridge/chamber and barrel features; record make, markings, serial number and action — 1 mark
- .315: commonly a rifle chambered for .315-inch-class ammunition in Indian usage; typically rifled barrel — 1 mark
- Document calibre/chamber markings, rifling, action, serial/maker marks and condition; do not rely on calibre label alone — 1 mark

16. Describe how the range of firing for a shotgun can be determined. (4 marks)

- Examine wound/target and distribution of pellets, wad and residue — 1 mark
- Conduct controlled test-fires using the same firearm/ammunition at known distances — 1 mark
- Compare pellet spread, wad impacts and GSR/soot patterns with questioned evidence — 1 mark
- Account for ammunition, barrel/choke, target, intervening material and environmental variation; estimate range rather than claim exact distance — 1 mark

17. Describe the Price test and its significance in firearm examination. (4 marks)

- A presumptive chemical test associated with detecting lead residues around a suspected bullet hole — 1 mark
- Reagent reaction/colour change is interpreted according to the prescribed method and controls — 1 mark

- May assist in locating/assessing bullet holes or residue patterns — 1 mark
- Not conclusive alone; possible false results and need for confirmatory/alternative examination — 1 mark

18. Define intermediate ballistics and explain its importance in the study of firearm projectile motion. (4 marks)

- Study of projectile behaviour as it exits the muzzle and interacts with gases near the firearm — 1 mark
- Covers transition from internal to external ballistics — 1 mark
- Includes muzzle blast/flash, gas effects and projectile disturbance near muzzle — 1 mark
- Helps understand stability, accuracy, trajectory and firing effects — 1 mark

19. Describe the magazine and firing mechanism of a firearm and explain their functions. (4 marks)

- Magazine stores and feeds cartridges into the firearm — 1 mark
- May be detachable or integral; spring/follower commonly assist feeding — 1 mark
- Firing mechanism includes trigger, sear and hammer/striker components — 1 mark
- Trigger releases the mechanism to strike the primer; safe handling and design vary by firearm — 1 mark

Part C – Essay Type Questions

Answer any 1 question. Each question carries 10 marks. Maximum: 10 marks.

20. Explain the significance of GSR composition, positioning and collection in establishing a connection between a person and a firearm incident. (10 marks)

- Define gunshot residue (GSR) and explain that it may contain primer/propellant and projectile-related particles — 2 marks
- Composition: characteristic inorganic/organic constituents and the importance of validated analytical methods — 2 marks
- Positioning: hands, clothing and nearby surfaces; distribution depends on distance, firearm, ammunition and activity — 2 marks
- Collection: timely collection, suitable validated kits, separate samples, contamination prevention, controls and chain of custody — 2 marks
- Interpretation: GSR supports an association but does not by itself prove firing; transfer, persistence, background and absence limitations — 2 marks

21. Discuss the forensic examination of improvised, country-made, and imitative firearms and the challenges associated with their identification. (10 marks)

- Safe receipt, documentation, photography, make/markings and condition — 2 marks
- Qualified examiner checks design, construction, barrel/chamber, action and safety/functionality without unsafe handling — 2 marks
- Test-firing only where lawful and safe, in a controlled facility; recover bullets/cartridge cases for comparison — 2 marks
- Assess calibre/ammunition compatibility, rifling/toolmarks, firing pin and breech-face marks where present — 2 marks
- Challenges: non-standard construction, defects, smoothbore/irregular barrels, altered marks, limited reference data and uncertain functionality; state limitations — 2 marks

Marking note: The paper instructs students to attempt at least one question from each course outcome. Apply the examination regulations when evaluating attempt requirements.